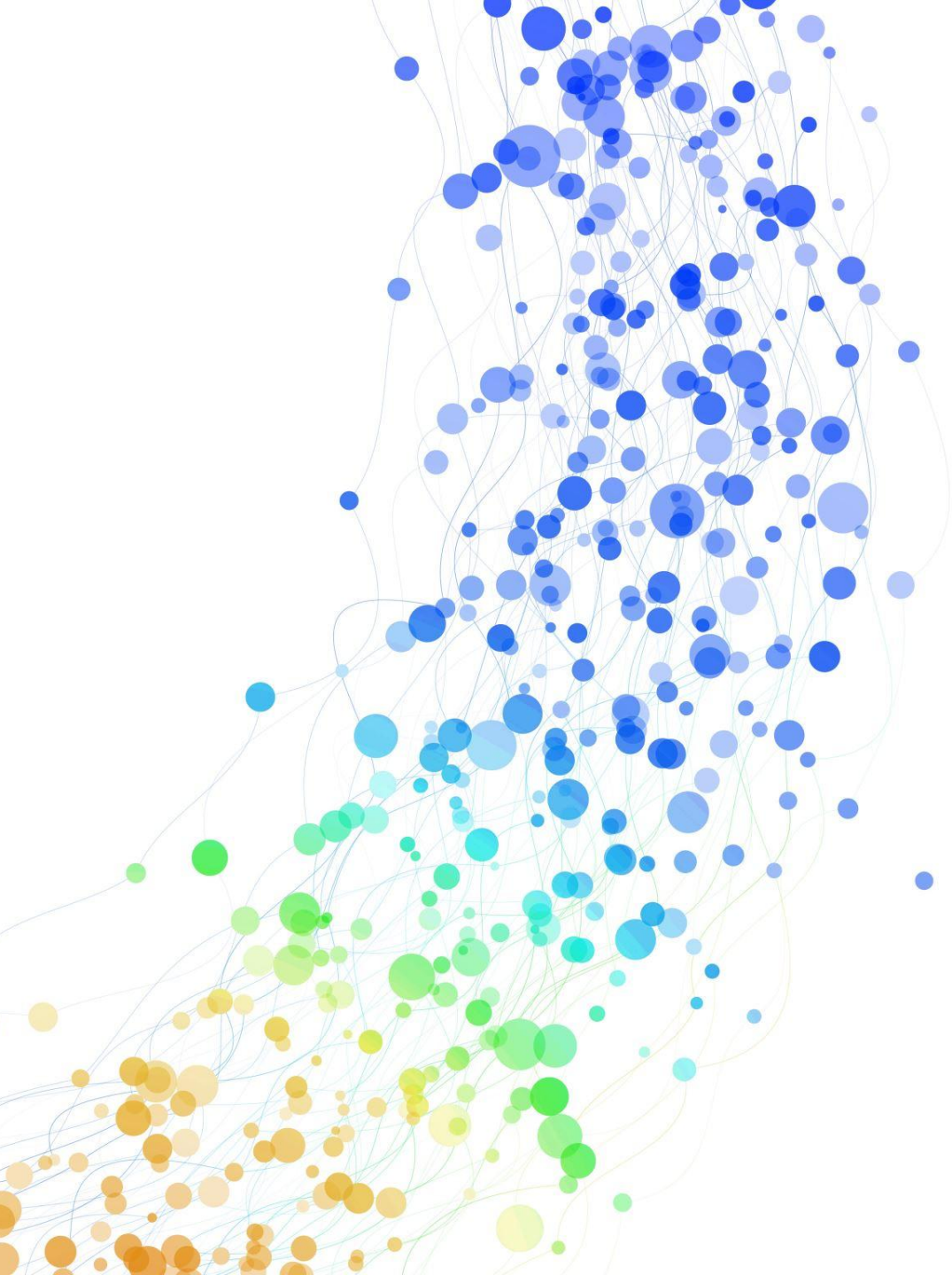

DengAI: Predicting Disease Spread

Members:
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Gabriela
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Introduction

Background:

- Dengue fever is a mosquito-borne disease prevalent in tropical and subtropical regions.
- Transmission dynamics are influenced by climate variables like temperature and precipitation.

Significance:

- Accurate prediction of dengue outbreaks aids public health initiatives and resource allocation.

+ Objective:

- Develop machine learning models to predict dengue outbreaks using climate and disease data.

First Dataset:

- Data from Zona da Mata, Minas Gerais, Brazil.
- Sources: INMET and DATASUS/SINAN.
- Issues: Low data quality and reliability, potential inadequate preprocessing

This score is calculated by Kaggle.

Completeness · 100%

- ✓ Subtitle
- ✓ Tag
- ✓ Description
- ✓ Cover Image

Credibility · 33%

- ✓ Source/Provenance
- ✗ Public Notebook
- ✗ Update Frequency

Compatibility · 75%

- ✓ License
- ✓ File Format
- ✓ File Description
- ✗ Column Description

Second Dataset (DengAI):

Second Dataset (DengAI):

- Weekly dengue case counts and environmental variables for San Juan (Puerto Rico) and Iquitos (Peru).
- There was a benchmark that had one strong issue which was it couldn’t predict peaks/outbreaks correctly

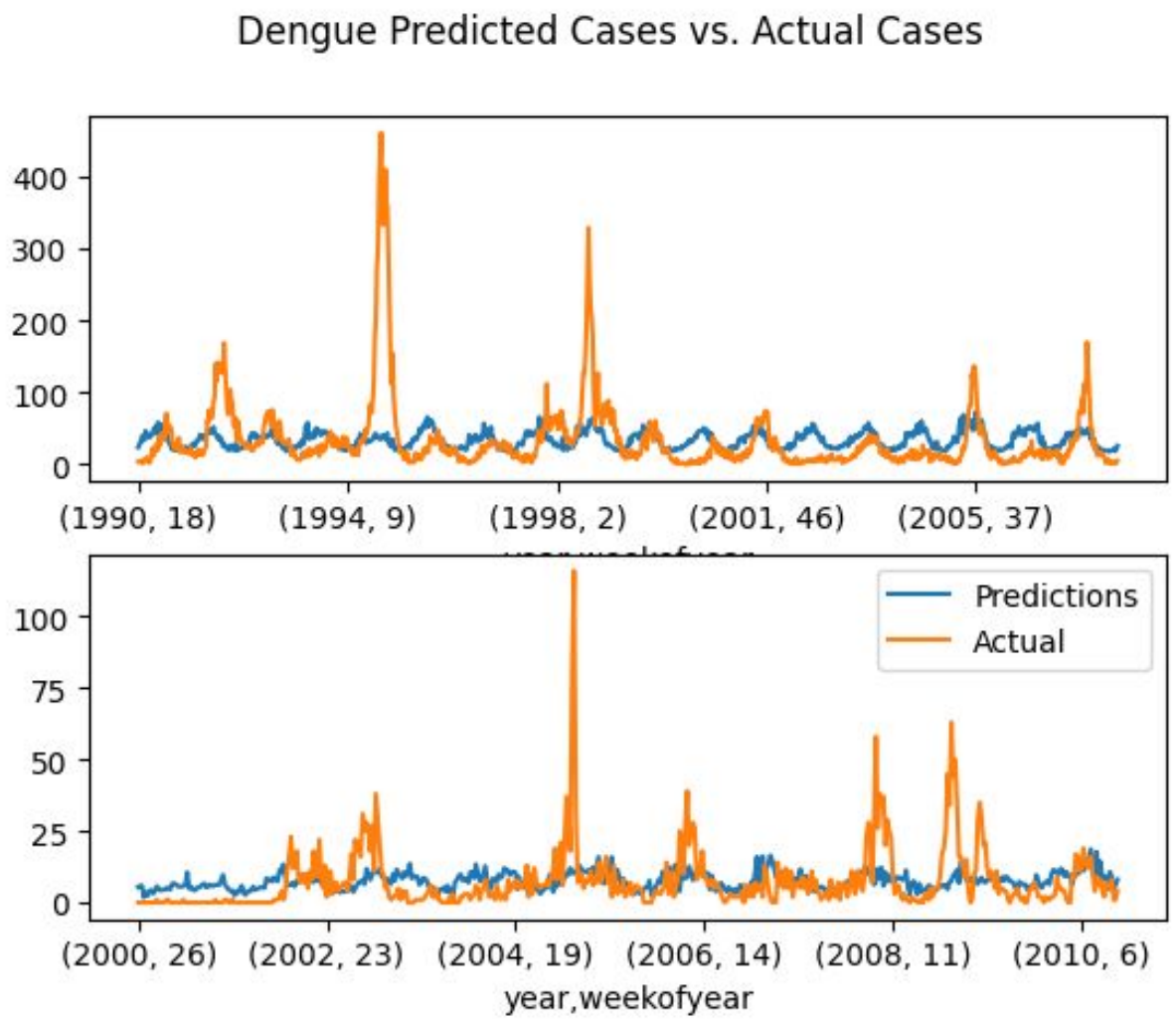
Benchmark Performance

San Juan

- MAE: 22.08
- R2: 0.14
- MSE: 756.98

Iquitos

- MAE: 6.47
- R2: 0.0036
- MSE: 119.32



First Approach

- The first approach was simple, swapping the model used for a different one
- The models used were KNN Regression, Decision Tree Regression, SVM Regression, and Random Forest Regression
- How the data was handled was equivalent to how the benchmark did apart from normalizing the features
- Hyperparameters were tuned using GridSearchCV
- In a way it was a way to compare the model they used to ones we knew about, as well as perhaps gain insight into we would need to improve the performance

First Approach Results

- Generally, all the models apart from one simply ended converging towards about the same result as the benchmark
- SVM was the only the model to perform better having a MAE of about 17.07
- Even then it still faced the same issue where the peaks/outbreaks weren't correctly predicted

Insights and Lead up to Second Approach

- It was obvious that for this particularly problem a simple model swap wouldn't necessarily improve performance
- This was further shown with the first approach as well as when simply adding the vegetation indexes 2 models got better performance than the benchmark
- Although the same issue resides where outbreaks aren't correctly predicted

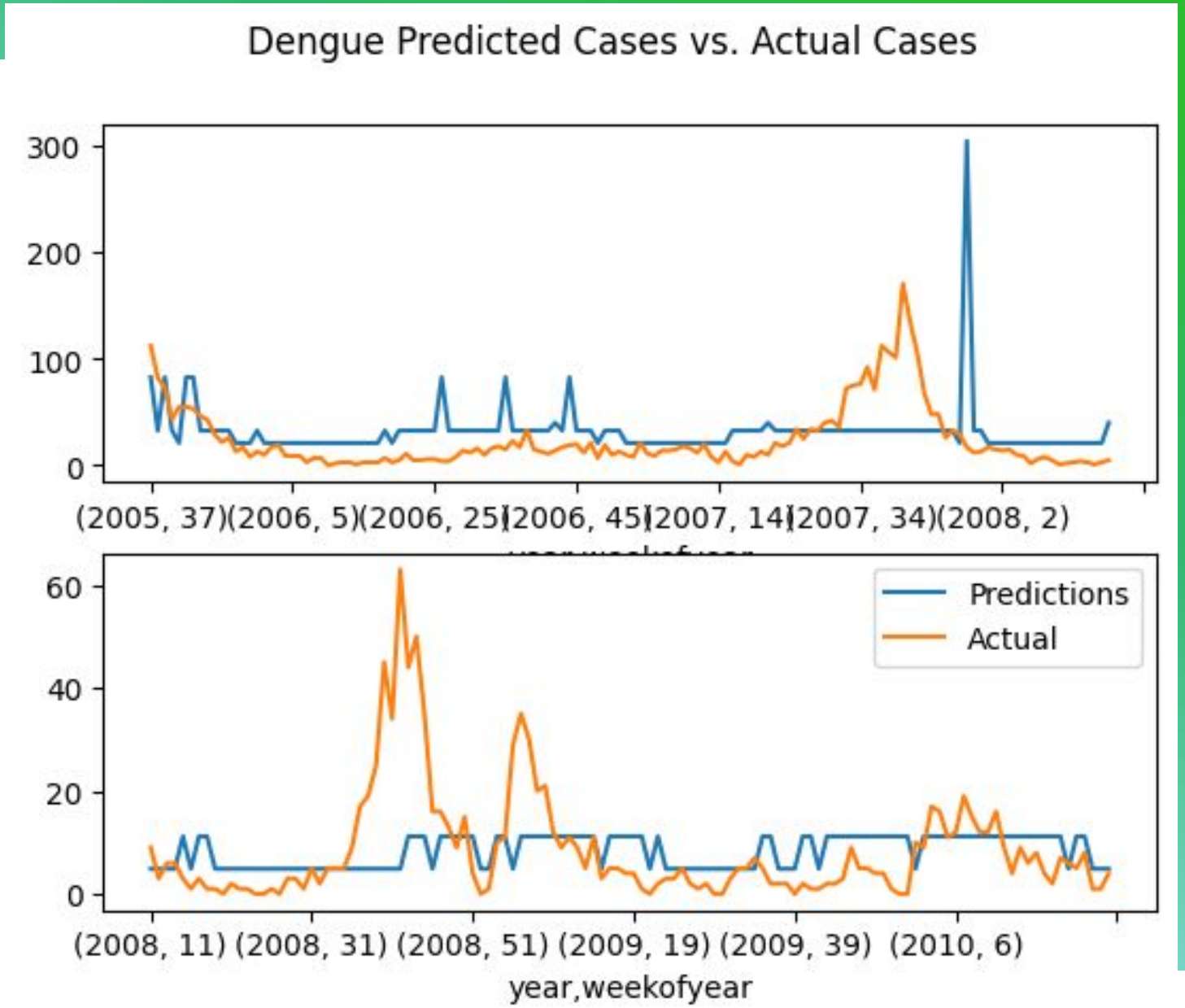
Decision Tree Regression

San Juan

- MAE: 23.03
- R2: -0.65
- MSE: 1446.34

Iquitos

- MAE: 6.78
- R2: 0.00062
- MSE: 119.67



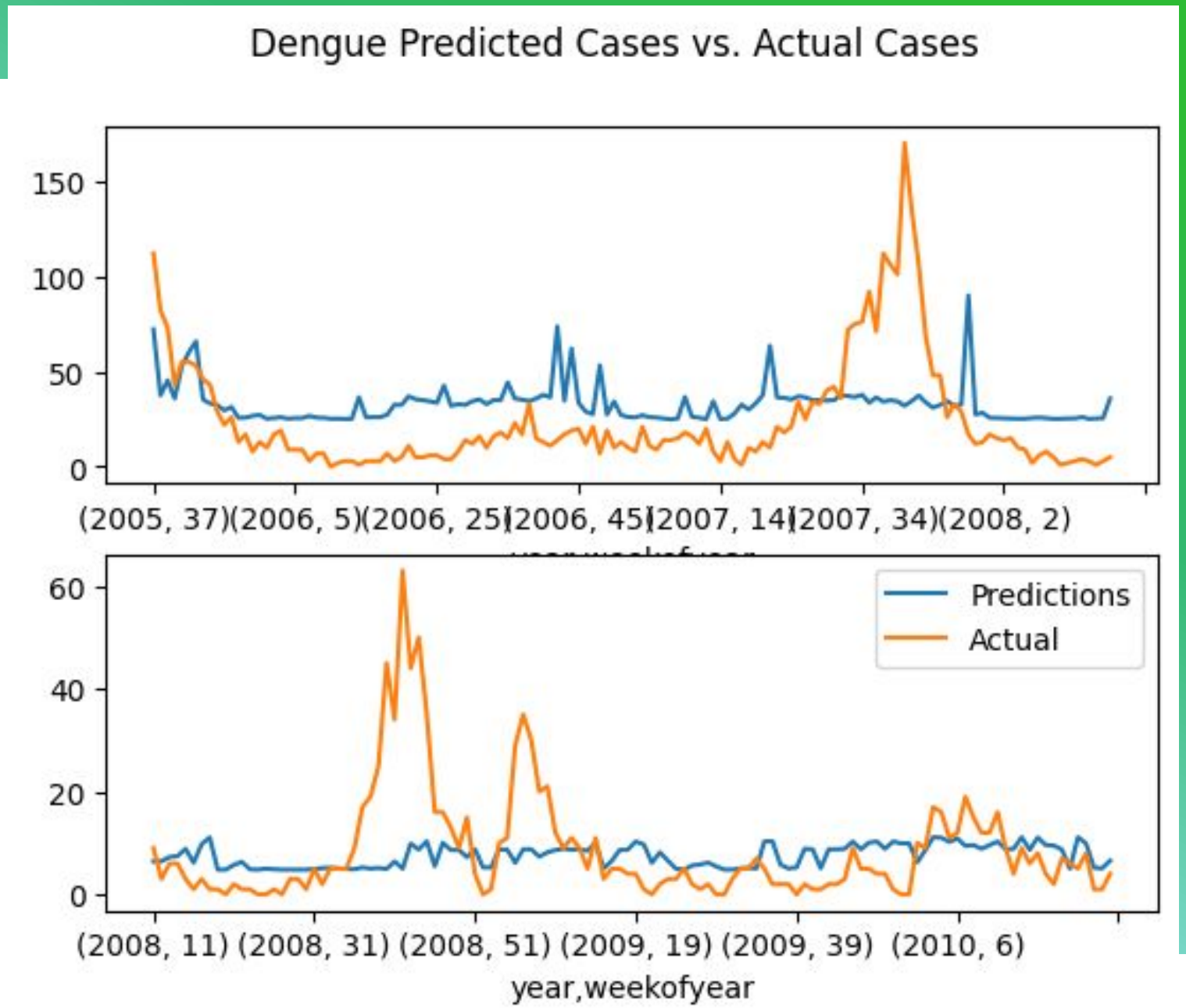
Random Forest Regression

San Juan

- MAE: 22.87
- R2: -0.01
- MSE: 884.73

Iquitos

- MAE: 6.61
- R2: 0.0034
- MSE: 119.34



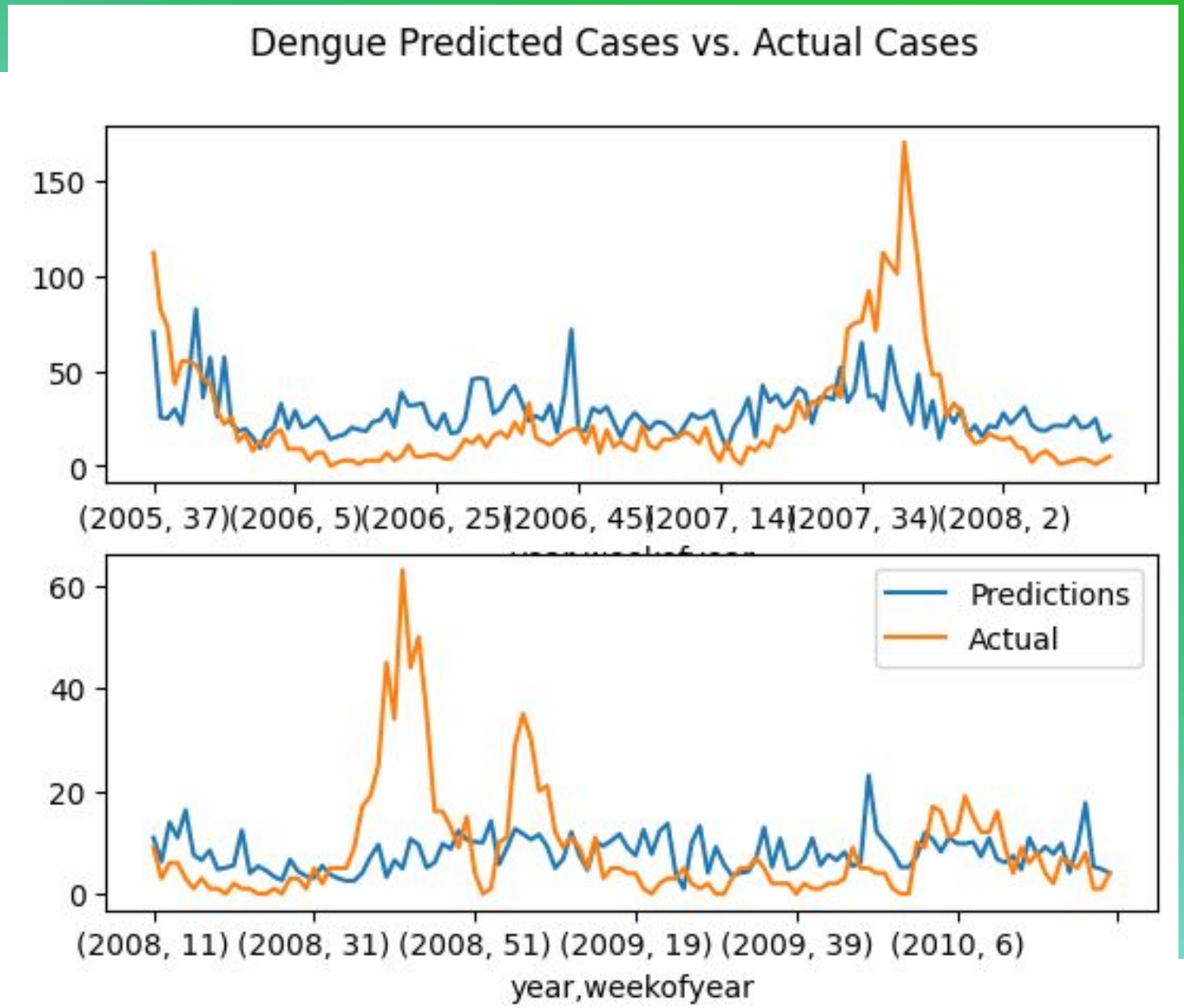
KNN Regression

San Juan

- MAE: 19.24
- R2: 0.17
- MSE: 728.90

Iquitos

- MAE: 7.2962
- R2: -0.061
- MSE: 127.06



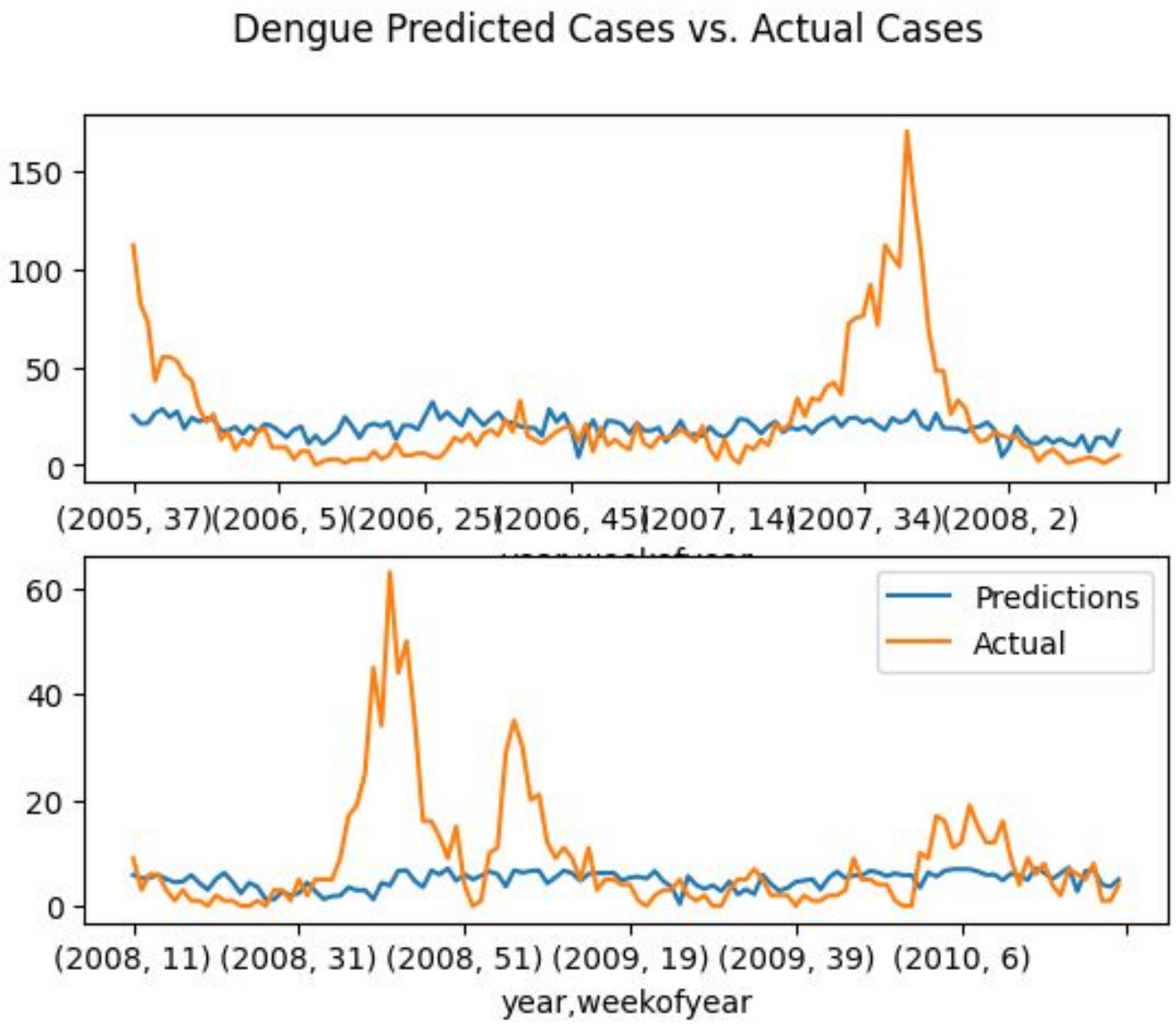
SVM Regression

San Juan

- MAE: 17.19
- R2: 0.06
- MSE: 823.64

Iquitos

- MAE: 6.49
- R2: -0.104
- MSE: 132.21



Second Approach:

Feature Engineering

NOAA's GHCN daily climate data weather station measurements

- station_max_temp_c – Maximum temperature
- station_min_temp_c – Minimum temperature
- station_avg_temp_c – Average temperature
- station_precip_mm – Total precipitation
- station_diur_temp_rng_c – Diurnal temperature range

PERSIANN satellite precipitation measurements (0.25x0.25 degree scale)

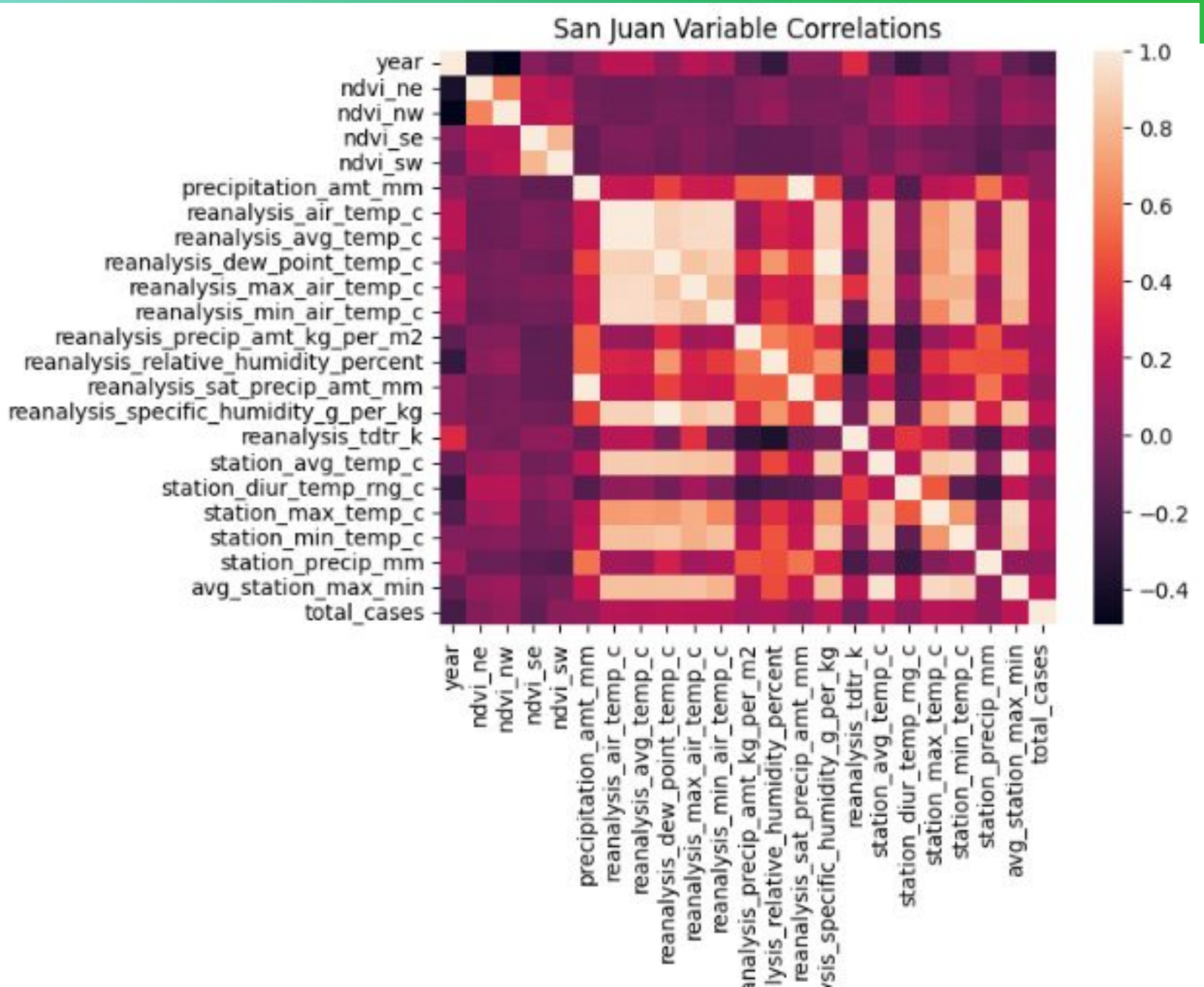
- precipitation_amt_mm – Total precipitation

NOAA's NCEP Climate Forecast System Reanalysis measurements (0.5x0.5 degree scale)

- reanalysis_sat_precip_amt_mm – Total precipitation
- reanalysis_dew_point_temp_k – Mean dew point temperature
- reanalysis_air_temp_k – Mean air temperature
- reanalysis_relative_humidity_percent – Mean relative humidity
- reanalysis_specific_humidity_g_per_kg – Mean specific humidity
- reanalysis_precip_amt_kg_per_m2 – Total precipitation
- reanalysis_max_air_temp_k – Maximum air temperature
- reanalysis_min_air_temp_k – Minimum air temperature
- reanalysis_avg_temp_k – Average air temperature
- reanalysis_tdtr_k – Diurnal temperature range

Satellite vegetation - Normalized difference vegetation index (NDVI) - NOAA's CDR Normalized Difference Vegetation Index (0.5x0.5 degree scale) measurements

- ndvi_se – Pixel southeast of city centroid
- ndvi_sw – Pixel southwest of city centroid


San Juan Variable Correlations
year
ndvi_ne
ndvi_nw
ndvi_se
ndvi_sw
precipitation_amt_mm
reanalysis_air_temp_c
reanalysis_avg_temp_c
reanalysis_dew_point_temp_c
reanalysis_max_air_temp_c
reanalysis_min_air_temp_c
reanalysis_precip_amt_kg_per_m2
reanalysis_relative_humidity_percent
reanalysis_sat_precip_amt_mm
reanalysis_specific_humidity_g_per_kg
reanalysis_tdtr_k
station_avg_temp_c
station_diur_temp_rng_c
station_max_temp_c
station_min_temp_c
station_precip_mm
avg_station_max_min
total_cases
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

```python
# Compute pairwise correlations for the updated dataset
correlation_matrix_updated = features_df_updated.drop(['city', 'week_start_date', 'weekofyear'], axis = 1).corr()
# Identify pairs of features with high correlation (absolute value > 0.8)
high_correlation_pairs = [
    (col1, col2, correlation_matrix_updated.loc[col1, col2])
    for col1 in correlation_matrix_updated.columns
    for col2 in correlation_matrix_updated.columns
    if col1 != col2 and abs(correlation_matrix_updated.loc[col1, col2]) > 0.8
]

# Identify pairs of features with high correlation (absolute value > 0.8)
high_correlation_pairs = [
    (col1, col2, correlation_matrix_updated.loc[col1, col2])
    for col1 in correlation_matrix_updated.columns
    for col2 in correlation_matrix_updated.columns
    if col1 != col2 and abs(correlation_matrix_updated.loc[col1, col2]) > 0.8
]

# Convert to a DataFrame for better visualization
redundant_features_df = pd.DataFrame(
    high_correlation_pairs, columns=["Feature 1", "Feature 2", "Correlation"]
)
redundant_features_df
```

Selected features and their relevance to dengue outbreaks

Remove Dummies Approach (NEXT)

	Feature 1	Feature 2	Correlation
0	ndvi_ne	ndvi_nw	0.850939
1	ndvi_nw	ndvi_ne	0.850939
2	ndvi_se	ndvi_sw	0.821125
3	ndvi_sw	ndvi_se	0.821125
4	precipitation_amt_mm	reanalysis_sat_precip_amt_mm	1.000000
5	reanalysis_dew_point_temp_c	reanalysis_specific_humidity_g_per_kg	0.997051
6	reanalysis_max_air_temp_c	reanalysis_tdtr_k	0.918579
7	reanalysis_max_air_temp_c	station_diur_temp_rng_c	0.834262
8	reanalysis_min_air_temp_c	reanalysis_tdtr_k	-0.815511
9	reanalysis_sat_precip_amt_mm	precipitation_amt_mm	1.000000
10	reanalysis_specific_humidity_g_per_kg	reanalysis_dew_point_temp_c	0.997051
11	reanalysis_tdtr_k	reanalysis_max_air_temp_c	0.918579
12	reanalysis_tdtr_k	reanalysis_min_air_temp_c	-0.815511
13	reanalysis_tdtr_k	station_diur_temp_rng_c	0.881175
14	station_diur_temp_rng_c	reanalysis_max_air_temp_c	0.834262
15	station_diur_temp_rng_c	reanalysis_tdtr_k	0.881175
16	station_max_temp_c	avg_station_max_min	0.814439
17	avg_station_max_min	station_max_temp_c	0.814439

```
[391] one_hot_sj = pd.get_dummies(X_sj['year'])
      X_sj = X_sj.drop('year', axis=1)
      X_sj = X_sj.join(one_hot_sj)
```

```
one_hot_iq = pd.get_dummies(X_iq['year'])
X_iq = X_iq.drop('year', axis=1)
X_iq = X_iq.join(one_hot_iq)
```

```
[393] years = X_sj.columns[:20:-1].values.tolist()
      X_sj = X_sj[years[::-1]].join(X_sj[X_sj.columns[:19]])
      X_sj.head()
```

	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	...	reanalysis_relative_humidity_percent	reanalysis_sat_precip_amt_mm	reanalysis_tdtr_k	station_avg_temp_c
0	False	False	False	False	False	False	False	False	False	False	...	73.366	12.42	2.629	25.443
1	False	False	False	False	False	False	False	False	False	False	...	77.369	22.82	2.371	26.714
2	False	False	False	False	False	False	False	False	False	False	...	82.053	34.54	2.300	26.714
3	False	False	False	False	False	False	False	False	False	False	...	80.337	15.36	2.429	27.471
4	False	False	False	False	False	False	False	False	False	False	...	80.460	7.52	3.014	28.943

rows × 35 columns

```
[394] years = X_iq.columns[:20:-1].values.tolist()
      X_iq = X_iq[years[::-1]].join(X_iq[X_iq.columns[:19]])
      X_iq.head()
```

	2003	2004	2005	2006	2007	2008	2009	2010	ndvi_ne	ndvi_nw	...	reanalysis_relative_humidity_percent	reanalysis_sat_precip_amt_mm	reanalysis_tdtr_k	station_av
936	False	False	False	False	False	False	False	False	0.193	0.132	...	92.419	25.41	8.929	
937	False	False	False	False	False	False	False	False	0.217	0.276	...	93.581	60.61	10.314	
938	False	False	False	False	False	False	False	False	0.177	0.173	...	95.849	55.52	7.386	
939	False	False	False	False	False	False	False	False	0.228	0.145	...	87.234	5.60	9.114	
940	False	False	False	False	False	False	False	False	0.329	0.322		88.161	62.76	9.500	

Second Approach Results

K-Neighbors Regressor

```
# sj knn Regressor
sj_knn = train_and_evaluate_knn(X_sj_train, X_sj_test, y_sj_train, y_sj_test)

# iq knn Regressor
iq_knn = train_and_evaluate_knn(X_iq_train, X_iq_test, y_iq_train, y_iq_test)
```

```
Mean Absolute Error: 19.582978723404256
Mean Absolute Error: 6.607692307692308
```

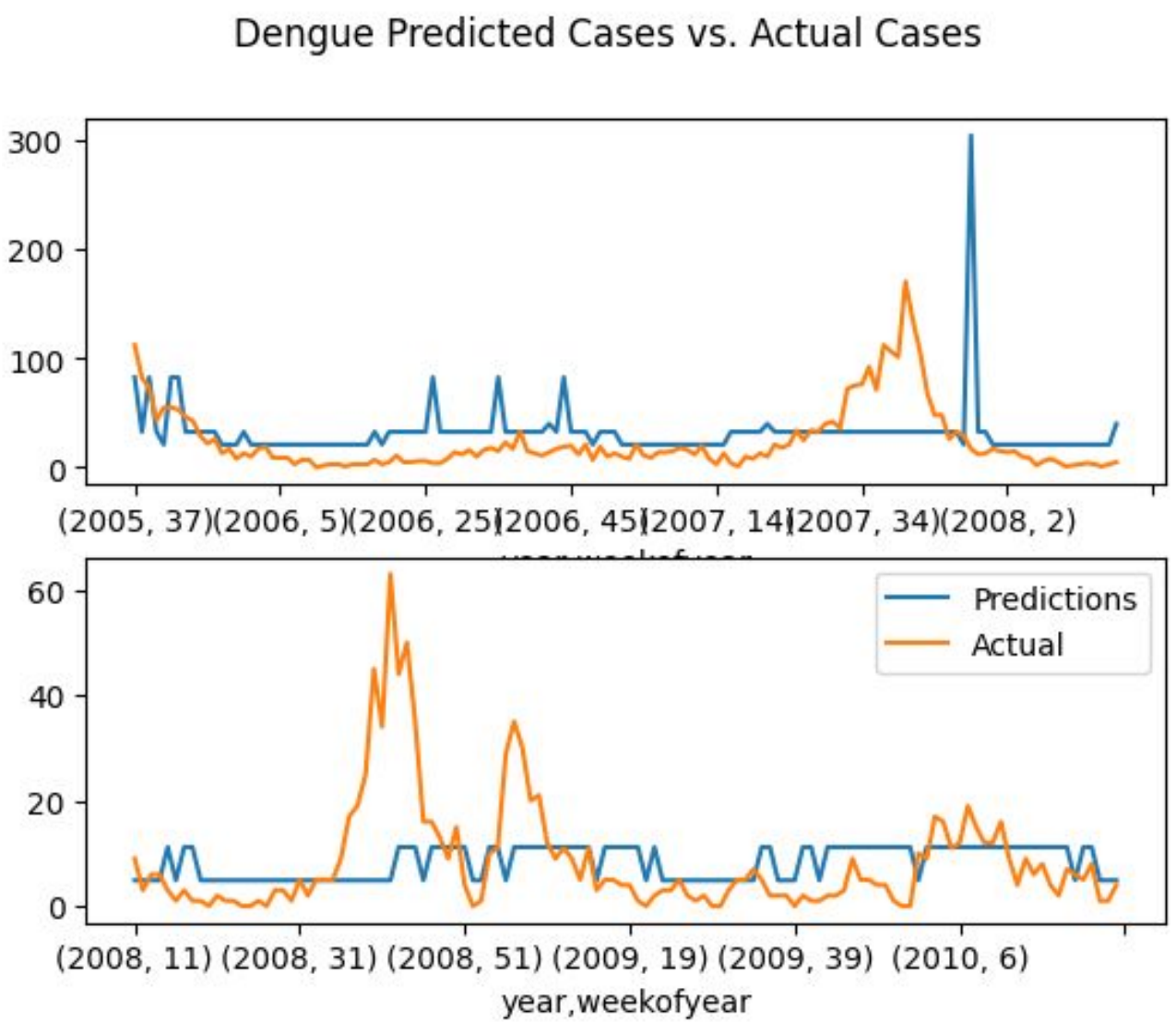
```
# sj rf
sj_rf = train_and_evaluate_rf(X_sj_train, X_sj_test, y_sj_train, y_sj_test)

# iq rf
iq_rf = train_and_evaluate_rf(X_iq_train, X_iq_test, y_iq_train, y_iq_test)
```

```
Mean Absolute Error: 20.107494064581854
Mean Absolute Error: 6.964110798914839
```

Random forest

Third Approach

- Taking a look back at the Decision Tree Regression Graph we noticed that it does almost predict a peak, but kind of delayed
- The non-improvement through feature selection and engineering showed perhaps we need a temporal factor and a model that could handle that



Third Approach

https://acrobat.adobe.com/id/urn:aaid:sc:VA6C2:91e682a0-ec6c-473d-bc74-7f7efa68d965